

◆ What is `ServletContext`?

`ServletContext` is a **global object** provided by the **Servlet container (like Tomcat)** that allows **all servlets in the application** to:

- Access shared information (like context parameters).
- Share data between servlets.
- Access server resources like files, logs, etc.

It is like the "application-level environment" shared across your servlets and JSPs.

◆ Key Features of `ServletContext`

Feature	Description
Scope	Application-wide (shared across all servlets)
Lifetime	Created when the application is deployed and destroyed when it's undeployed
Access Method	<code>getServletContext()</code> from <code>HttpServlet</code>
Defined Parameters	In <code>web.xml</code> using <code><context-param></code>
Communication	Allows inter-servlet communication via attributes

◆ When and Why is `ServletContext` Needed?

✓ **When you want:**

- To store shared configuration values (like company name, db URL, etc.).
- To make files/resources available across multiple servlets.
- To **share objects** (e.g., a database connection pool, counter) between servlets.

◆ What is `ServletConfig`?

`ServletConfig` is an **interface** in the `javax.servlet` package used to **pass configuration information** to a **specific servlet** at initialization time.

It acts like a **configuration object** that is **unique per servlet**.

□ Why Do We Need `ServletConfig`?

When you have a servlet that requires **initialization parameters** (like DB connection settings, API keys, etc.), you don't want to hardcode those in the Java class. Instead, you can specify them in `web.xml`, and the servlet can access them using `ServletConfig`.

✓ Key Features

- Exists **per servlet**
 - Used to read **init parameters** declared in `web.xml`
 - Allows **parameterization** of servlet behavior
 - Retrieved using `getServletConfig()`
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✂ Key Methods in `ServletConfig`

Method	Description
<code>getInitParameter(String name)</code>	Returns the value of the given init param
<code>getInitParameterNames()</code>	Returns all init param names as Enumeration
<code>getServletName()</code>	Returns the name of the servlet
<code>getServletContext()</code>	Returns a reference to <code>ServletContext</code>

Feature	ServletContext	ServletConfig
Definition	Object used to communicate between the servlet and the container for the entire web application	Object used by the servlet container to pass initialization parameters to a particular servlet
Scope	Application-wide scope – shared across all servlets	Servlet-specific scope – available only to that servlet
Lifetime	Exists as long as the web application is running	Exists as long as the servlet instance is alive
Use Case	Storing shared objects like DB connections, app-level settings	Configuring specific servlet parameters like file paths, limits, etc.
Access Method	getServletContext() (from GenericServlet or HttpServlet)	getServletConfig() (from GenericServlet or HttpServlet)
Initialization Source	Defined in web.xml or dynamically managed via code or annotations	Usually defined in web.xml under <servlet><init-param>
Typical Usage	- Application-level parameters- Logging utilities- Counters	- Servlet-specific configs- Servlet init values
Sharing Between Servlets	Yes (shared among all servlets)	No (not shared; local to that servlet)

🎯 Problem Statement:

Design a web-based **Book Info Display System** using **Java Servlets**, where:

- `ServletConfig` is used by an admin to configure **per-servlet details** like `storeName`, `adminEmail`, or `bookCategory`.
 - `ServletContext` is used to **share and update global visitor statistics** across all pages (like total visits, active users, and site-wide settings).
 - Multiple users can visit different book categories (fiction, science, technology) and the system keeps track of total and per-category visits.
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📋 Functional Requirements:

1. **AdminConfigServlet:**

- Uses `ServletConfig` to display servlet-specific initialization parameters (e.g., store name, admin email, category name).
- Reads config like:
 - `storeName: BookVerse`
 - `adminEmail: admin@bookverse.com`
 - `category: Fiction`

2. **BookCategoryServlet:**

- Uses `ServletContext` to:
 - Track total visitors to the site (`totalVisitors`)
 - Track visitors per category (`fictionCount`, `techCount`, `scienceCount`)
 - Show global application version (from context param)

3. **LandingServlet:**

- A default welcome servlet that increments the total visitor count and shows a list of categories.

Sample Output:

🔗 *Landing Page (/welcome)*

Welcome to BookVerse!

Total Visitors So Far: 27

Select a Category:

1. Fiction
2. Science
3. Technology

🔗 *Fiction Category Servlet Output:*

Book Category: Fiction

Store: BookVerse

Admin Email: admin@bookverse.com

Fiction Books Visited: 12

Total Visitors: 27

App Version: 1.2.3

🔗 *Admin Configuration Servlet Output:*

Admin Panel

Configured Category: Science

Store Name: BookVerse

Contact Admin: admin@bookverse.com